

Unit 1B Group Problem Session

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Cal Poly CSC 3445, Fall 2026

1 Instructions

Solve these problems together, in a group of four.

Each team member will write up the solution to one of the questions.

Submit your solutions to Gradescope (more instructions to follow in Canvas).

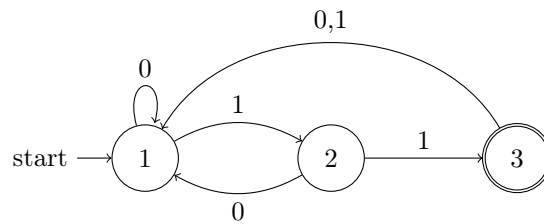
You will need to provide:

- The names of all team members
- Your solutions to the problems, indicating who wrote up each solution.
- **You may consult generative AI tools and other sources, but do not use them to write your solutions.**
- Cite any external sources or tools used and you used them, or indicate that you did not use any. **This part is required.**

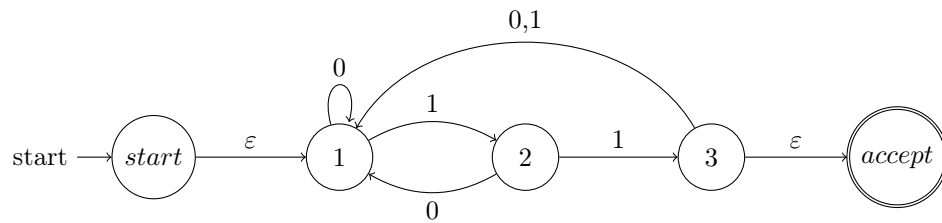
2 Questions

1. Consider the procedure for converting a DFA to an equivalent regular expression, from class.

Here is the initial machine:



After adding a new start and accept state, the machine is as follows:

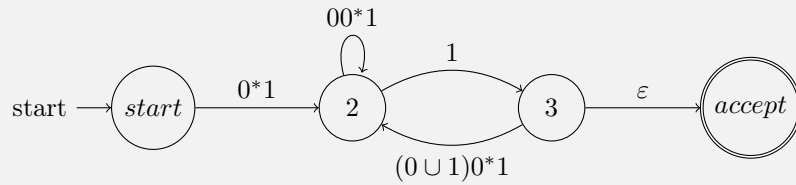


Draw the machine that results from eliminating state 1, including the regular expression(s) appearing on each new transition. Then eliminate state 2, then state 3; then write the resulting regular expression. Also write each intermediate machine.

Written by: Emilio Meza

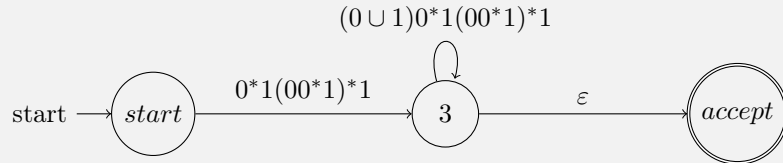
Step 1: Eliminate state 1. State 1 has a self-loop labeled 0, so each new transition gets 0^* in the middle. The transitions into state 1 come from *start* (label ε), from 2 (label 0), and from 3 (label $0 \cup 1$). The only transition out of state 1 goes to 2 (label 1). This gives:

- $start \rightarrow 2$: $\varepsilon 0^*1 = 0^*1$
- $2 \rightarrow 2$ (new self-loop): $00^*1 = 00^*1$
- $3 \rightarrow 2$: $(0 \cup 1)0^*1$

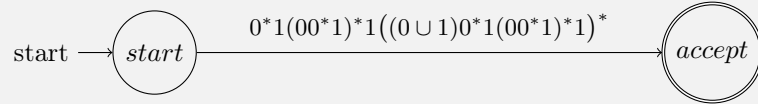


Step 2: Eliminate state 2. State 2 now has the self-loop 00^*1 , so each new transition gets $(00^*1)^*$ in the middle. The transitions into state 2 come from *start* (label 0^*1) and from 3 (label $(0 \cup 1)0^*1$). The only transition out of state 2 goes to 3 (label 1). This gives:

- $start \rightarrow 3$: $0^*1(00^*1)^*1$
- $3 \rightarrow 3$ (new self-loop): $(0 \cup 1)0^*1(00^*1)^*1$



Step 3: Eliminate state 3. The only path left is $start \rightarrow 3 \rightarrow accept$, and state 3 has a self-loop. The new label is (label into 3)(self-loop)*(label out of 3). The trailing ε does not change the expression.

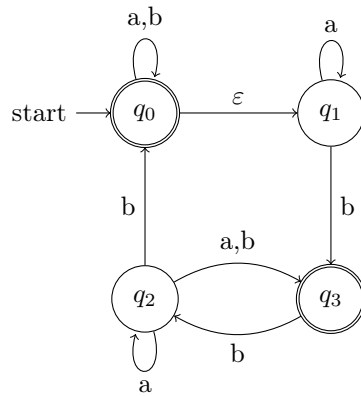


Resulting regular expression:

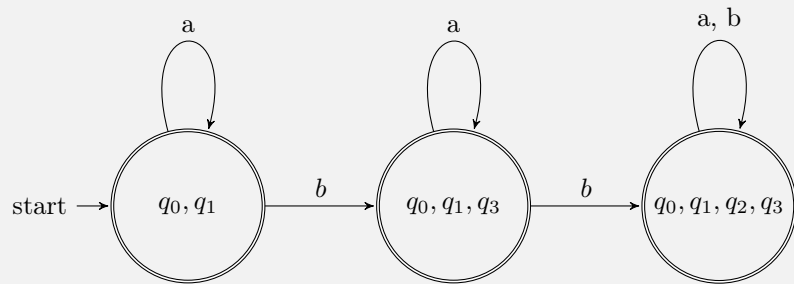
$$0^*1(00^*1)^*1((0 \cup 1)0^*1(00^*1)^*1)^*$$

Sources: Used Claude (Anthropic AI) to help work through the state-elimination steps and typeset this solution.

2. Convert the following NFA to an equivalent DFA, using the procedure from class. You do not have to draw unreachable or dead states.



Written by: Chris Amey



Sources: Two youtube videos were used for reference and as a refresher on conversions. They can be found [here](#) and [here](#). No AI tools were used at any point in research, determining the solution, and writing the solution.

3. Let $\Sigma = \{a, b, c\}$. Prove that the language

$$L_1 = \{c^l a^m b^n \mid m \geq 2n, n \geq 1, l \geq 0\}$$

is not regular using the Pumping Lemma.

For reference, here is the statement of the Pumping Lemma:

Lemma (Pumping Lemma): Let L be a regular language. Then there exists some number p (the “pumping length”) such that the following holds:

Let w be a string in L where $|w| \geq p$. Then there exist substrings x, y, z where $w = xyz$ and:

- (a) For every $i \geq 0$, $xy^i z \in L$.
- (b) $|y| > 0$.
- (c) $|xy| \leq p$.

Give your proof that L_1 is not regular below. (When referring to conditions 1-3 in your proof, you do not need to rewrite the statement of each condition. You can just say "conditions 1-3 hold" at the relevant place in your proof.)

Proof:

Written by: Robert Cruz-Diaz

Suppose for a contradiction that L_1 is regular.

Then $\exists$ pumping length, p .

Let $w = c^{0p} a^{2p} b^p \Rightarrow w = a^{2p} b^p$. $w \in L_1$ & $|w| \geq p$

Then by Pumping lemma, $w = xyz$ where

$$(3) \quad \forall i \geq 0, xy^i z \in L$$

$$(3) \quad |y| > 0$$

$$(3) \quad |xy| \leq p$$

Let $w' = xy^0 z \Rightarrow w' = xz$.

By (3), xy contains only a 's since $|xy| \leq p$ and that the first p symbols of w are a 's. By (2), y contains at least one a .

$\therefore w' = xz$ has fewer than $2p$ a 's & still has p b 's. Hence $w' \notin L_1$, contradicting (1).

Sources: Went to Office hours. Did not need to use any AI tools to reach this solution.

4. Let $\Sigma = \{a, b, c\}$. Prove that the language

$$L = \{a^m b^n \mid m \geq 2n, n \geq 0\}$$

is not regular, using one or more closure properties of regular languages and the fact that L_1 from the previous problem is not regular.

(Hint: you may need to use concatenation and find two additional regular languages, say L_2 and L_3 .)

Written by: Omar Garcia

Proof. Suppose for a contradiction that L is regular.

Let $L_1 = \{c^l a^m b^n \mid m \geq 2n, n \geq 1, l \geq 0\}$. Let $L_2 = c^*$. Let $L_3 = \Sigma^* b \Sigma^*$.

$$L_1 = L_3 \cap (L_2 \cdot L)$$

Looking at L_1 , we see that in order to reach L_1 , we are in need of concatenating a string $c^l \in L_2$ that satisfies a form $c^l a^m b^n$. By concatenating $L \cdot L_2$, we get:

$$(L \cdot L_2) = \{c^l a^m b^n \mid m \geq 2n, n \geq 0, l \geq 0\}.$$

In order to reach L_1 , $n \geq 1$. Right now our string allows for 0 b 's, but L_1 says we need at least 1 b . So intersecting with L_3 gives us strings with at least 1 b .

Hence:

$$L_3 \cap (L \cdot L_2) = \{c^l a^m b^n \mid m \geq 2n, l \geq 0, n \geq 1\} = L_1.$$

We assumed L is regular, L_2 is regular, and by closure under concatenation, $L \cdot L_2$ is regular. L_3 is regular, and again by closure under intersection, $L_3 \cap (L \cdot L_2)$ is regular. So L_1 is regular.

However, here we have a **contradiction**. We know from the previous problem that L_1 is **NOT** regular. In the proof above, we claim L_1 is regular. Thus we have a contradiction.

Since assuming L is regular led to L_1 being regular, which is false, L must not be regular. $\square$

Sources: Used lecture videos on youtube. No AI needed.